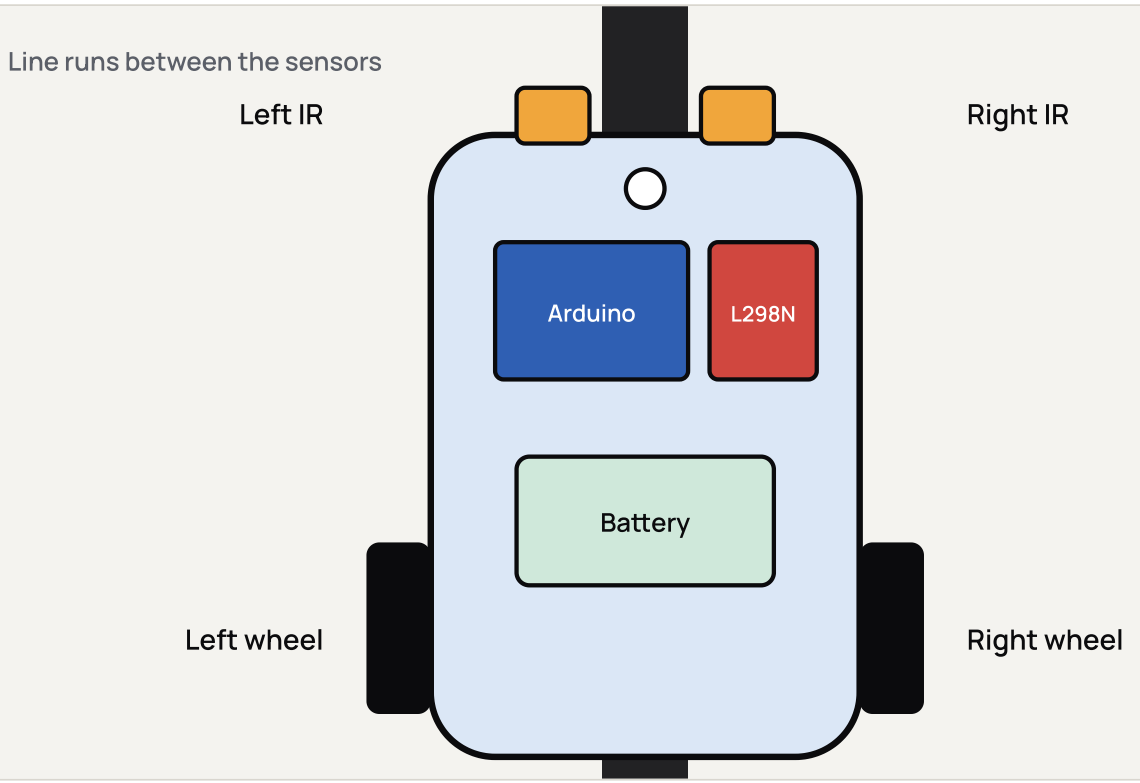
Assembly guide

Step-by-step build

Safety rules

- Ask an adult before using batteries. Use only the battery pack from the kit.
- Never let the battery's red and black wires touch each other.
- Switch off and unplug power before changing any wire.
- Keep fingers, hair and loose clothes away from spinning wheels.
- If anything gets hot, smells or smokes: switch off and tell an adult.



Assembly guide

1. **Lay out every part.** Check everything on the kit list is there before you start. Keep screws in a small bowl.
2. **Fix the two motors.** Screw each motor to the underside of the chassis at the back, shafts pointing outwards. The wires should face inwards.
   Tip: Tighten until firm, not as hard as you can. Plastic brackets crack.
3. **Push on the wheels.** Line up the flat side of the shaft with the wheel's hole and press straight on.
4. **Mount the castor.** Fix the castor wheel under the front so the chassis sits level.
5. **Fix the sensors.** Mount both IR modules at the front edge, facing down, about 5 to 10 mm above the floor and about 3 cm apart, one either side of where the line will be.
   Tip: Your black tape should fit between the two sensors with a little gap on each side.

6. **Mount the Arduino and driver.** Fix the Arduino Uno on spacers on top of the chassis, with the USB port reachable. Fix the L298N next to it.
7. **Fit the battery holder.** Fix the holder where it balances the robot, usually over the back wheels. Do not put batteries in yet.

Build check

- [ ] Both motors are firmly fixed and do not wobble
- [ ] Both wheels spin freely by hand without rubbing
- [ ] The robot sits level on its wheels and castor
- [ ] Both IR sensors face down, 5 to 10 mm above the floor
- [ ] The Arduino's USB port can be reached
- [ ] No batteries are in the holder yet

Adult wiring check (before power)

- [ ] Left IR OUT to D2, right IR OUT to D3
- [ ] Both IR VCC to 5V and GND to GND
- [ ] ENA to D5, ENB to D6 (jumper caps removed)
- [ ] IN1 D8, IN2 D9, IN3 D10, IN4 D11
- [ ] L298N GND connected to Arduino GND (common ground)
- [ ] Battery + to L298N 12V/VS through the switch, battery − to L298N GND
- [ ] No motor wire goes to any Arduino pin
- [ ] No bare wire ends touching each other